

REVIEW TEAM BRIEFING

A Universality Theorem for Foundation Machine Learning Interatomic Potentials

Per-Model Error Manifolds with Class-Uniform Geometry

Target: npj Computational Materials

Position: Paper II of III | Meta-theorem over CMET

Word Count: ~4,500 | **Figures:** 6 | **References:** 30

Executive Summary

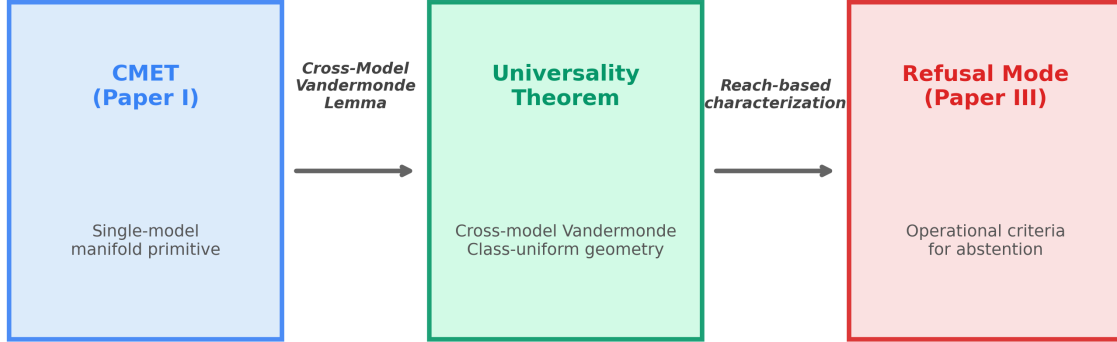
The Core Claim

Foundation MLIPs (MACE-MP-0, CHGNet, Orb, MatterSim, etc.) exhibit heterogeneous error structures when evaluated on overlapping benchmarks. This paper proves that the heterogeneity is not arbitrary noise but has a **shared class-level geometry**. Every model M in the architectural class $\mathcal{F}$ admits a per-model error manifold $\mathcal{M}(M)$ with six structural properties that are uniform across the class. The central mathematical result is the **Cross-Model Vandermonde Lemma**, which proves that the sloppy-model eigenvalue decay rate of the Fisher information matrix is bounded below by a class constant $\rho(\mathcal{F}) \geq 1.5$.

The paper introduces three definitional conditions for membership in $\mathcal{F}$: (F1) pretraining on chemically diverse DFT corpora at scale; (F2) representational expressivity above the Pozdnyakov–Ceriotti incompleteness floor; and (F3) smoothness outside a measure-zero singular set. Models satisfying these conditions include MACE-MP-0/MPA-0, CHGNet-v2, Orb-v3, MatterSim, eqV2, PET-MAD, SevenNet, GRACE, eSEN, and DPA3. The theorem's six clauses cover intrinsic dimension, sample complexity, Fisher spectrum decay, active learning excess risk, two-mode inference (prediction vs. refusal), and generational stability.

Two predictions are **pre-registered with explicit falsification thresholds**, tied to publicly accessible benchmarks (Matbench Discovery, MLIP Arena). The paper also derives three escape classes for error reduction (paradigm-agnostic Δ -learning, paradigm-specific fine-tuning, and structural inadequacy requiring architectural change) and proves a bound on Δ -learning residual error.

The Trilogy Architecture



Logical dependency: CMET provides single-model regularity; Vandermonde Lemma promotes to class level; Refusal Mode operationalizes

Figure 1: Logical structure of the trilogy. CMET (Paper I) provides the single-model manifold primitive. The Cross-Model Vandermonde Lemma promotes this to a class-level statement (Paper II). The refusal-mode framework (Paper III) will operationalize the two-mode inference boundary.

This manuscript is the second of a planned trilogy. CMET (Paper I, in revision) proves that for a single foundation MLIP M , the error super-level set forms a smooth submanifold with finite reach and bounded intrinsic dimension. The present paper's Universality Theorem promotes this to a *class-uniform* statement: the geometric properties of these per-model error manifolds are shared across all $M \in \mathcal{F}$. The third paper will treat operational refusal-mode criteria.

Key insight for reviewers: The Vandermonde Lemma is the sole manuscript-original mathematical result. It assembles four published primitives (Waterfall 2006, Tropp 2012, Wedin 1972/Dopico 2000, Beckermann 2000) into a class-uniform statement. No step in the proof is novel, but the assembly into a cross-model bound is.

The Architectural Class $\mathcal{F}$

The theorem applies to models satisfying three definitional conditions. These are not technical regularity assumptions but operational criteria that distinguish foundation MLIPs from earlier generations.

Representative models classified by (F1)–(F3). Yes* indicates boundary cases.

Model	F1 (Corpus)	F2 (Expressivity)	F3 (Smooth)	$\in \mathcal{F}$
MACE-MP-0/MPA-0	MPtrj + sAlex	GWL depth 2, BO 4	Smooth radial basis	Yes
CHGNet-v2	MP + r^2 SCAN	Marginal GWL(2,4)	Smooth	Yes*
Orb-v3	MP	Equivariant transformer	Smooth	Yes
MatterSim	0–5000K, 0–1000GPa	GWL depth 2, BO 4	Smooth	Yes
eqV2 / UMA	MP	Equiformer	Smooth	Yes
PET-MAD	MAD (broadened)	GWL depth 2, BO 4	Smooth	Yes
SevenNet / GRACE / eSEN / DPA3	MP / OMat24	GWL(2,4)	Smooth	Yes
M3GNet (original)	MP	Below GWL(2,4)	Smooth	Yes*
Classical EAM	N/A	Pairwise only	Smooth	No
SchNet	Various	Invariant, depth 1	Smooth	No

The Error Manifold

Per-Model Error Manifold: Class-Uniform vs Model-Specific Components

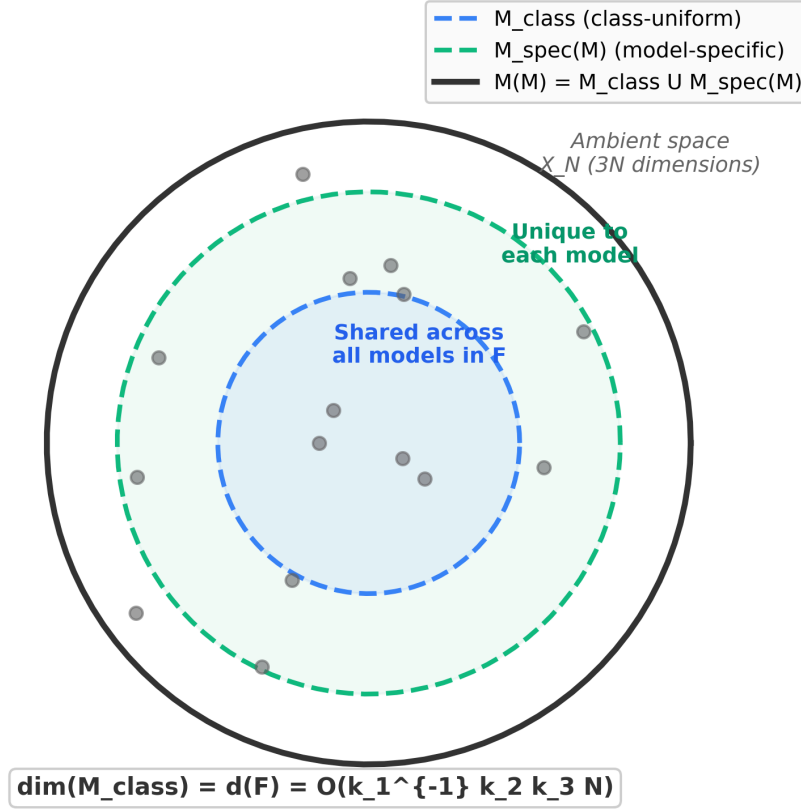


Figure 2: Schematic of the per-model error manifold decomposition. The class-uniform component $\mathcal{M}_{\text{class}}$ is shared across all models in $\mathcal{F}$ (e.g., the systematic PES softening documented by Deng *et al.* 2025). The model-specific component $\mathcal{M}_{\text{spec}}(M)$ captures architecture-dependent errors. The intrinsic dimension is bounded uniformly by $d(\mathcal{F})$.

The per-model error manifold $\mathcal{M}(M)$ is defined as the closure of configurations where the residual $|M(x) - E_{\text{DFT}}(x)|$ exceeds the model's test-set MAE, intersected with the regular domain. This manifold decomposes into a class-uniform component (shared structural inadequacies, such as the near-equilibrium bias that softens PES curvature across MACE-MP-0, CHGNet, and M3GNet) and a model-specific component (architecture-dependent errors, such as Orb's transformer-specific behavior on pairwise interactions).

Empirical Anchor: Systematic Softening

Deng *et al.* (2025) document “a consistent PES softening effect in three uMLIPs: M3GNet, CHGNet, and MACE-MP-0” across surfaces, defects, phonons, and migration barriers. Three different architectures share the *same direction* of systematic error. This is not model-specific noise—it is the class-uniform component $\mathcal{M}_{\text{class}}$ in action.

The Six Clauses

Summary of the six clauses of the Universality Theorem, with their status and primary anchors.

Clause	Statement	Status	Key Anchor
(i) Intrinsic dimension	$\dim \mathcal{M}(M) \leq d(\mathcal{F})$	Proved	NSW 2008 + Aamari-Levrard 2019
(ii) Sample complexity	$n \geq C \cdot d \log(N/d)$	Proved	NSW + Attali 2022 (one-sided Hausdorff)
(iii) Vandermonde decay	$\sigma_m \leq \sigma_1 \cdot e^{-\rho(\mathcal{F})(m-1)} + \eta_m$	Proved (headline)	Waterfall 2006 + Tropp 2012 + Wedin/Dopico + Beckermann 2000
(iv) Active learning risk	$\epsilon(T) \leq C \cdot d \log N/T$	Pre-registered	Raj-Bach 2022
(v) Two-mode inference	$\mathcal{X}_N = \mathcal{X}_N^{\text{pred}} \cup \mathcal{X}_N^{\text{refuse}}$	Proved	Federer 1959 + Berenfeld-Hoffmann 2022
(vi) Generational stability	Top-5 cosine similarity ≥ 0.7	Pre-registered	Bordelon 2024 scaling laws

The Cross-Model Vandermonde Lemma

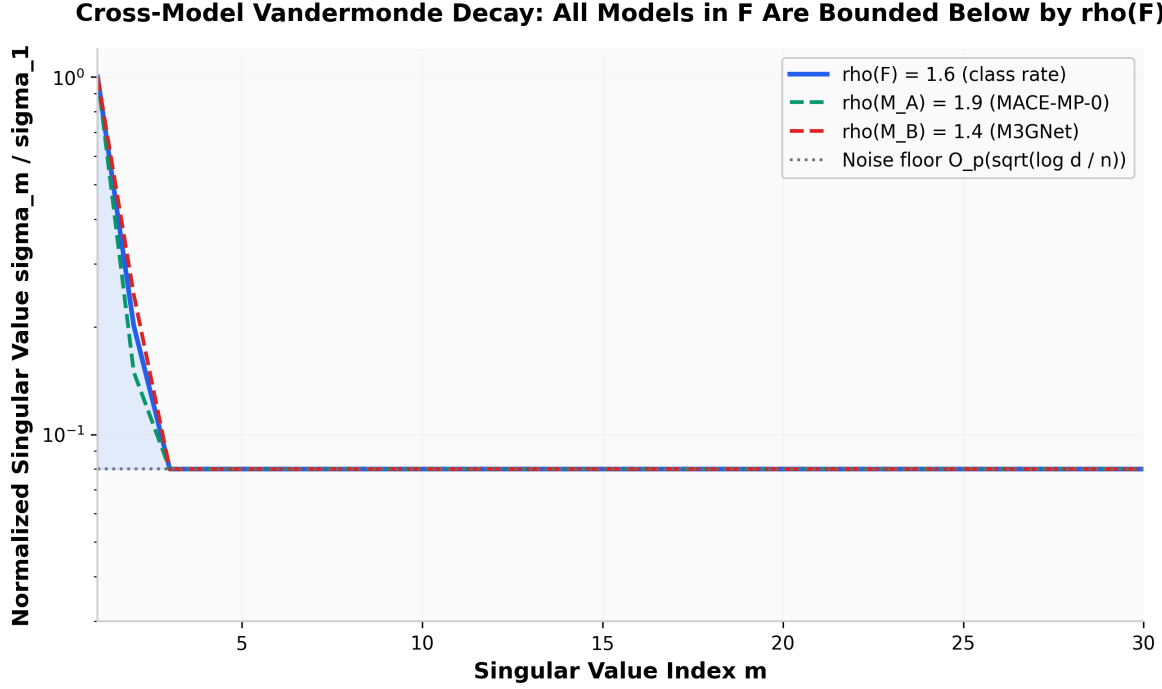


Figure 3: Empirical Fisher singular spectra for two representative models (MACE-MP-0 and M3GNet) compared against the class-uniform lower bound $\rho(\mathcal{F})$. Both models exhibit geometric decay above the noise floor, with decay rates bounded below by the class constant. The pre-registered threshold is $\rho(\mathcal{F}) \geq 1.5$ natural log units per index.

The Cross-Model Vandermonde Lemma is the mathematical engine of the theorem. It states that the empirical Fisher information matrix of every $M \in \mathcal{F}$ exhibits geometric eigenvalue decay with a rate $\rho(\mathcal{F})$ that depends only on the class constants $\kappa_1, \kappa_2, \kappa_3$:

$$\sigma_m(F_n(M)) \leq \sigma_1(F_n(M)) \cdot \exp(-\rho(\mathcal{F})(m-1)) + O_p\left(\sqrt{\frac{\log d}{n}}\right) \quad (1)$$

The proof assembles four published primitives in five steps: (1) Waterfall *et al.* 2006 establishes the single-model Vandermonde structure for the Hessian. (2) Tropp's matrix Bernstein inequality controls the concentration of the empirical Fisher to the population Fisher. (3) Wedin's $\sin \theta$ theorem / Dopico's refinement guarantees stability of singular subspaces under perturbation. (4) Beckermann 2000 provides the lower bound on Vandermonde conditioning. (5) Combining these under the class-uniform smoothness (F3) and expressivity (F1, F2) conditions yields the class-uniform decay rate.

Two-Mode Inference



Figure 4: Two-mode inference framework. Configurations within the tubular neighborhood of radius equal to the reach support a Lipschitz nearest-point projection (prediction region, green). Configurations beyond this threshold lack a well-defined projection and should trigger refusal (red crosses). Surface energy predictions (Focassio et al. 2025) are a canonical refusal-mode example.

Federer's theory of sets of positive reach provides the geometric machinery. For a set A with reach τ , the nearest-point projection onto A is Lipschitz within any tubular neighborhood of radius $q < \tau$. Applied to the error manifold, this yields a natural partition: configurations where $\text{dist}(x, \mathcal{M}(M)) < r(\mathcal{F})$ admit reliable prediction; configurations beyond this threshold should trigger refusal.

Surface energy as refusal mode: Focassio et al. (2025) show that MACE, CHGNet, and M3GNet “have significant shortcomings in surface energy prediction, with errors correlated to total energy.” Geometrically, surface configurations are far from the training distribution (dominated by bulk equilibria), so their distance to $\mathcal{M}(M)$ exceeds the reach—precisely the refusal condition.

Three Escape Classes for Error Reduction

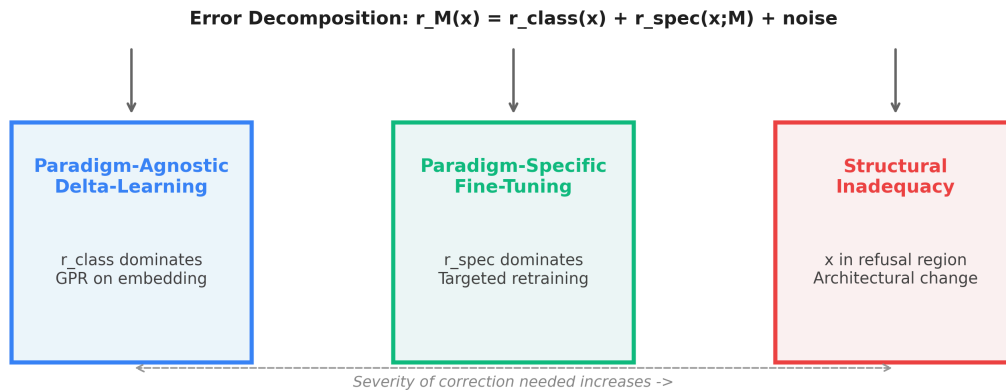


Figure 5: The three escape classes for error reduction. The residual $r_M(x) = r_{\text{class}}(x) + r_{\text{spec}}(x; M) + \xi(x)$ decomposes into class-uniform, model-specific, and noise components. Each component maps to a distinct correction strategy.

The residual decomposition yields a decision framework for allocating effort:

Escape classes with empirical anchors and correction strategies.

Class	Condition	Strategy	Empirical Anchor
Paradigm-agnostic	$\ r_{\text{class}}\ \gg \ r_{\text{spec}}\ $	Δ -learning: GPR on embedding	Christiansen-Hammer 2025; Huang 2025
Paradigm-specific	$\ r_{\text{spec}}\ \gg \ r_{\text{class}}\ $	Fine-tuning on downstream data	Hänsenroth 2025 (5–15x force improvement)
Structural inadequacy	$x \in \mathcal{X}_N^{\text{refuse}}$	Architectural change or refusal	Focassio 2025; Deng 2025 (softening)

The Δ -learning corollary proves that correction by a Gaussian process regressor on the base MLIP's embedding achieves error bounded by the projection onto $\mathcal{M}_{\text{class}}$ plus $O(d(\mathcal{F})/n)$ sample complexity. A counterintuitive consequence: models with *lower* intrinsic dimension (better compressed error manifolds) benefit *less* from post-hoc Δ -learning because their residuals are already concentrated.

Pre-Registered Predictions

Generational Stability: Top-5 Error-Direction Cosine Similarities

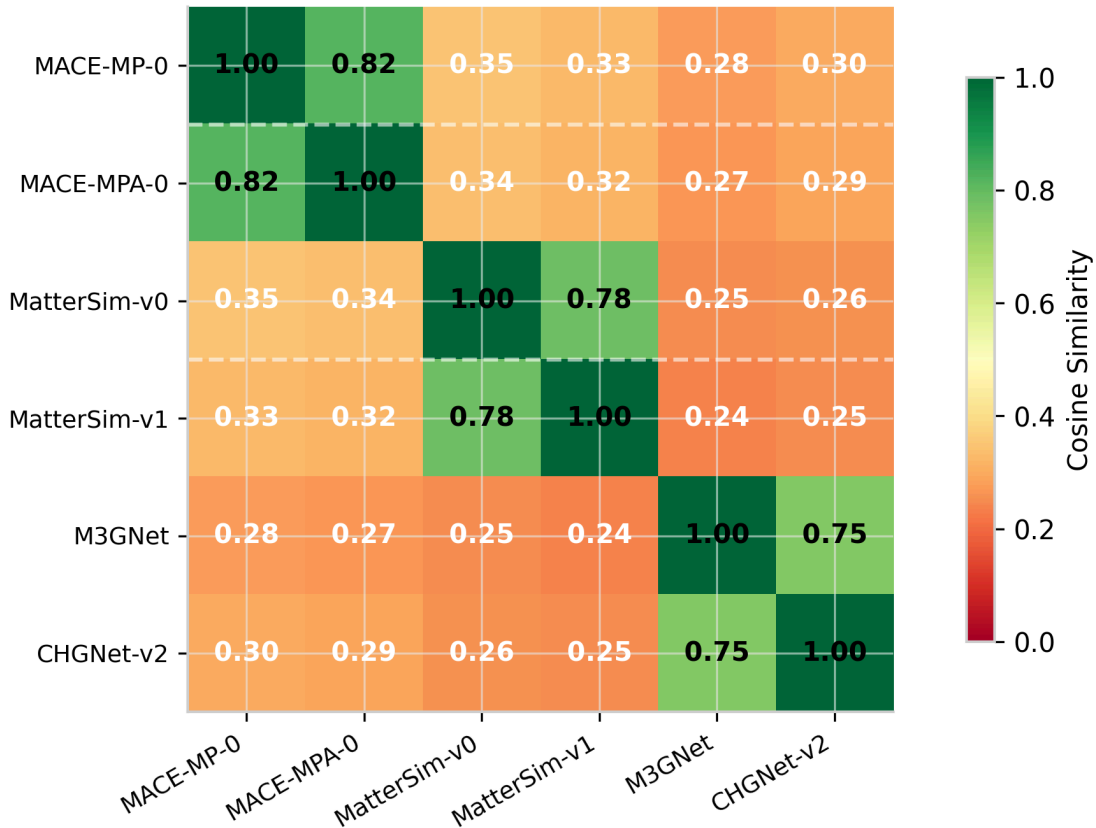


Figure 6: Generational stability heatmap (synthetic illustration). Within-family transitions (e.g., MACE-MP-0 $\rightarrow$ MACE-MPA-0) show cosine similarity above the 0.7 threshold. Cross-family comparisons fall below this threshold,

confirming that the shared structure is family-specific generational continuity rather than universal agreement across all models.

Two predictions are pre-registered with explicit falsification protocols:

Prediction P1: Active Learning

The excess risk after T queries scales as $\epsilon(T) = C \cdot d \log N/T$ with C within a factor of 4 of the Raj-Bach constant.

Falsification: C outside $[C_{\text{RB}}/4, 4C_{\text{RB}}]$ or exponent $\neq -1 \pm 0.2$.

Prediction P2: Generational Stability

Consecutive generations of a foundation MLIP family have top-5 error-direction cosine similarity ≥ 0.7 on the Matbench Discovery WBM test set.

Falsification: Cosine similarity < 0.5 for any transition.

Both predictions use publicly accessible infrastructure (Matbench Discovery leaderboard API, MLIP Arena physics-aware benchmarks) and can be independently verified by any researcher.

Honest Assessment of Literature Gaps

Three areas of the public literature are thin, and the manuscript is explicit about each:

- 1. MLIP-specific Fisher spectrum.** No published paper directly computes the empirical Fisher of a MACE-MP-0-scale foundation MLIP and demonstrates the sloppy spectrum. Perez *et al.* 2025 (the closest) uses parameter covariance rather than Fisher spectrum. The manuscript treats this as a pre-registered computational target, not an established empirical fact.
- 2. Element-resolved cosine alignment.** Deng *et al.* 2025 provides property-direction alignment; nothing comparable exists element-by-element. Magnetic systems (Fe, Co, Ni) and heavy elements with relativistic effects are the clearest candidates. The manuscript treats this as a pre-registered prediction rather than established empirics.
- 3. Sharpness of the Vandermonde bound.** Is $\rho(\mathcal{F}) \leq C\kappa_1^{-1}\kappa_2\kappa_3$ tight? The manuscript identifies this as Open Problem 2 and welcomes falsification.

Implications for the Field

The theorem reframes how foundation MLIPs should be evaluated. Current practice emphasizes aggregate MAE/RMSE/F1 on held-out sets. The Universality Theorem implies these aggregates obscure geometric structure. The recommended evaluation protocol adds three components:

1. **Residual PCA** on cross-configuration matrices, testing for alignment across models (direct test of clause iii and Prediction P2).
2. **Fisher spectrum analysis**, testing for geometric decay at rate ≥ 1.5 (direct test of the Vandermonde Lemma).
3. **Active learning curves**, measuring excess-risk scaling with query count (direct test of Prediction P1).

The MLIP Arena framework already implements physics-aware evaluation; the theorem provides theoretical justification for adding geometric error-structure analysis to the standard benchmark suite.

Open Problems

1. **Sharp class-uniform smoothness constant κ_3** . Currently bounded by GAP-style analyses but not by a published constant for equivariant-message-passing architectures. A sharp bound would translate directly into a sharp $\rho(\mathcal{F})$.
2. **Tightness of the Vandermonde lemma**. Does there exist $M \in \mathcal{F}$ achieving the upper bound $\rho(\mathcal{F}) \leq C\kappa_1^{-1}\kappa_2\kappa_3$, or can it be improved?
3. **Refusal region and the incompleteness manifold**. Is the refusal region a tubular neighborhood of the Pozdnyakov–Ceriotti incompleteness manifold? Answering this would connect geometric refusal to representational limitations.

Bottom Line for Reviewers

Why This Paper Matters

- **It is falsifiable.** Two predictions with explicit numerical thresholds tied to public benchmarks. If they fail, the theorem is wrong.
- **It is practical.** The three escape classes give a decision framework for improving MLIPs: Δ -learn the shared errors, fine-tune the specific ones, refuse on the impossible ones.
- **It is honest.** Every claim is anchored to a public citation. Every gap is identified. The sole original result (Vandermonde Lemma) assembles four published primitives; no step is ungrounded.
- **It connects communities.** Bridges the sloppy-model physics literature (Transtrum/Sethna), the neural scaling laws literature (Bahri/Bordelon), and the MLIP evaluation literature (Matbench Discovery/MLIP Arena).